ATOM-SG Baseline Assessment Test

Primary 6 Mathematics — Geometry Track (25 Questions)

Name: _____ **Class:** _____ **Date:** _____

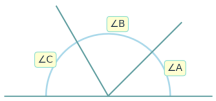
Instructions:

- This test contains 25 geometry questions. Time allowed: 45 minutes.
- Show all your working in the spaces provided. Marks may be awarded for correct method.
- Write your final answers clearly in the boxes provided.
- Calculators are NOT allowed unless specified.
- Diagrams are NOT drawn to scale unless stated otherwise.

Q 1. [A] In the diagram, measure ∠AOB, ∠BOC, and ∠COD using a protractor. Write your answers in the boxes provided.

(G1: Identify and measure angles using a protractor)

G001: Measure angles with protractor



Measure angles ∠A, ∠B, and ∠C in the diagram

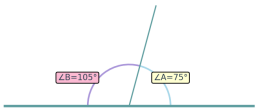
Working:

Answer: ∠AOB = _____°, ∠BOC = _____°, ∠COD = _____° **(3m)**

Q 2. [B] In the diagram, AOB is a straight line. If ∠AOC = 75°, find ∠COB.

(G1: Use angle properties (angles on straight line = 180°))

G002: Angles on straight line



Two angles on a straight line

Working:

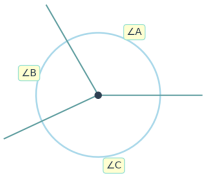
Answer: ∠COB = _____° **(2m)**

Baseline Assessment Test — Continued

Q 3. [B] Three angles meet at point O. Given ∠AOB = 120° and ∠BOC = 85°, find ∠AOC.

(G1: Use angle properties (angles at a point = 360°))

G003: Angles around a point



Three angles meet at a point

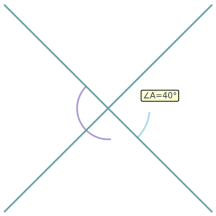
Working:

Answer: ∠AOC = _____° **(2m)**

Q 4. [B] In the diagram, two straight lines AB and CD intersect at O. If ∠AOC = 40°, find ∠AOD, ∠BOD, and ∠BOC.

(G1: Use angle properties (vertically opposite angles))

G004: Vertically opposite angles



Two straight lines intersect

Working:

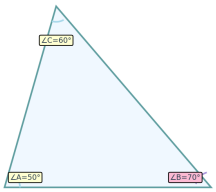
Answer: ∠AOD = _____°, ∠BOD = _____°, ∠BOC = _____° **(3m)**

Baseline Assessment Test — Continued

Q 5. [C] In triangle ABC, ∠A = 50° and ∠B = 70°. Find ∠C.

(G1: Apply angle sum of triangle (180°))

G005: Triangle angle sum



Triangle ABC with two angles given

Working:

Answer: ∠C = _____° **(2m)**

Q 6. [A] In the diagram, identify and label all pairs of perpendicular lines and parallel lines using the symbols ⊥ and ■.

(G1: Identify perpendicular and parallel lines)

[Diagram not found: G006-parallel-perpendicular-v1.svg]

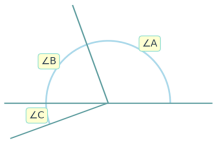
Working:

Answer: Perpendicular: _____, Parallel: _____ **(2m)**

Q 7. [D] In the diagram, AOB is a straight line. ∠BOD = 90° (right angle). If ∠AOC = 110°, find ∠COD.

(G1: Combined angle properties (multi-step))

G007: Combined angle properties



Two-step angle reasoning

Working:

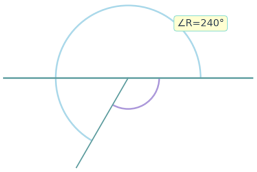
Answer: ∠COD = _____° **(3m)**

Baseline Assessment Test — Continued

Q 8. [C] Measure the reflex angle ∠XOY in the diagram (the angle greater than 180°).

(G1: Measure reflex angles)

G008: Reflex angle measurement



Measure reflex angle ∠R

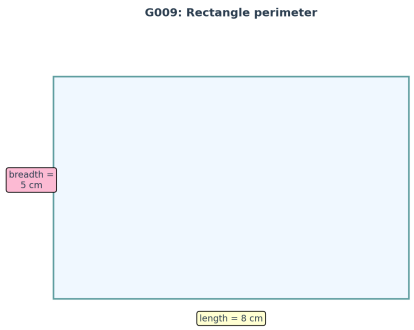
Working:

Answer: ∠XOY = _____° **(2m)**

Baseline Assessment Test — Page 2

Q 9. [A] Find the perimeter of rectangle ABCD with length 8 cm and breadth 5 cm.

(G2: Calculate perimeter of rectangles)

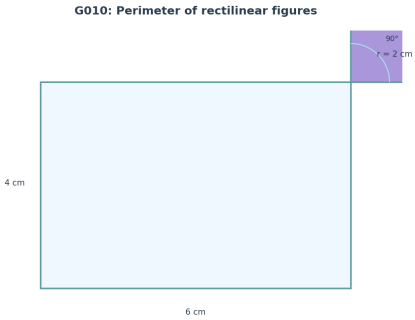


Working:

Answer: Perimeter = _____ cm **(2m)**

Q10. [B] The figure shows a composite shape made of two rectangles. Find its perimeter.

(G2: Calculate perimeter of composite rectilinear figures)



Working:

Answer: Perimeter = _____ cm **(3m)**

Baseline Assessment Test — Continued

Q11. [B] Find the area of the L-shaped figure shown. All dimensions are in centimetres.

(G2: Calculate area of composite rectangular figures)

[Diagram not found: G011-lshape-area-v1.svg]

Working:

Answer: Area = _____ cm^2 **(3m)**

Q12. [C] A rectangular sheet measures 12 cm by 8 cm. A 3 cm by 3 cm square is cut from one corner. Find the remaining area.

(G2: Calculate area with missing sections)

[Diagram not found: G012-cutout-rectangle-v1.svg]

Working:

Answer: Remaining area = _____ cm^2 **(3m)**

Q13. [A] Convert: (a) 3.5 m = _____ cm, (b) 250 cm = _____ m, (c) 4.2 km = _____ m

(G2: Convert between units of length)

Working:

Answer: (a) _____, (b) _____, (c) _____ **(3m)**

Q14. [B] Convert 2.5 m^2 to cm^2.

(G2: Convert between units of area)

Working:

Answer: 2.5 m^2 = _____ cm^2 **(2m)**

Baseline Assessment Test — Continued

Q15. [A] Draw all lines of symmetry on the regular hexagon shown.

(G4: Identify lines of symmetry)

[Diagram not found: G015-hexagon-symmetry-v1.svg]

Working:

Answer: Number of lines = _____ **(2m)**

Q16. [C] The figure shows half of a symmetrical shape with the mirror line (dotted). Complete the symmetrical figure.

(G4: Complete symmetrical figures)

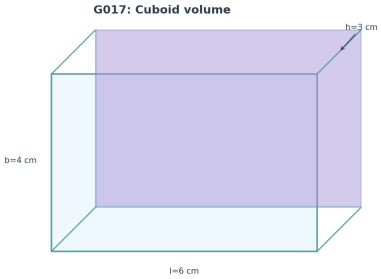
[Diagram not found: G016-symmetry-complete-v1.svg]

Working:

Answer: (Complete the drawing on the diagram) **(3m)**

Q17. [A] Find the volume of the cuboid with length 6 cm, breadth 4 cm, and height 3 cm.

(G3: Calculate volume of cuboids)



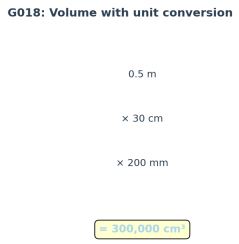
Working:

Answer: Volume = _____ cm^3 **(2m)**

Baseline Assessment Test — Page 3

Q18. [B] Find the volume of a box with dimensions 0.5 m × 30 cm × 200 mm. Give your answer in cm³.

(G3: Calculate volume with unit conversion)



Working:

Answer: Volume = _____ cm³ **(3m)**

Q19. [C] Which of the following nets can be folded to form a cube? Circle the correct answer(s).

(G3: Identify valid nets of cubes)

[Diagram not found: G019-cube-nets-v1.svg]

Working:

Answer: Valid net(s): _____ **(2m)**

Q20. [D] Draw a net of the cuboid shown on the grid. The dimensions are 4 cm × 2 cm × 1 cm.

(G3: Draw nets of cuboids)

[Diagram not found: G020-cuboid-net-v1.svg]

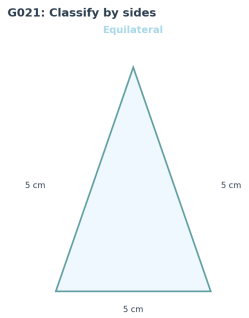
Working:

Answer: (Draw on the grid provided) **(4m)**

Baseline Assessment Test — Continued

Q21. [A] Classify the triangle with side lengths 5 cm, 5 cm, and 5 cm.

(G4: Classify triangles by sides)



Working:

Answer: Type: _____ triangle **(1m)**

Q22. [A] Classify the triangle with angles 45°, 45°, and 90°.

(G4: Classify triangles by angles)

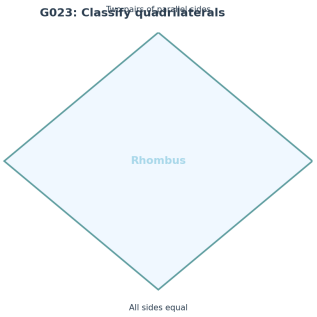
[Diagram not found: G022-triangle-angles-v1.svg]

Working:

Answer: Type: _____ triangle **(1m)**

Q23. [B] Identify the quadrilateral: It has two pairs of parallel sides, all sides equal, but angles are not 90°.

(G4: Classify quadrilaterals)



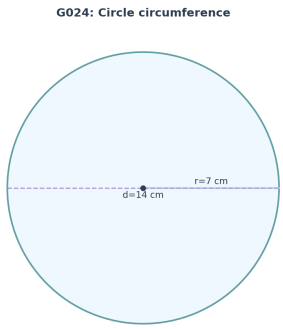
Working:

Answer: Quadrilateral: _____ **(2m)**

Baseline Assessment Test — Continued

Q24. [C] A circle has radius 7 cm. Find its circumference. (Take $\pi = 22/7$)

(G2: Calculate circumference of circle)

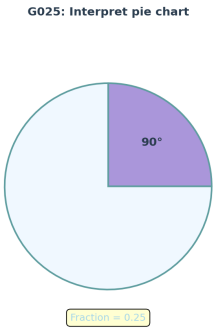


Working:

Answer: Circumference = _____ cm **(2m)**

Q25. [D] The pie chart shows how a student spends 24 hours in a day. If the 'Sleep' sector has angle 90°, what fraction of the day is spent sleeping?

(G2: Interpret pie charts as fractions)



Working:

Answer: Fraction = _____ **(3m)**

Answer Key — For Markers Only

DO NOT DISTRIBUTE TO STUDENTS

Q#	Answer	Subpathway	Marks
Q 1	∠AOB = _____°, ∠BOC = _____°, ∠COD ...	G1: Identify and measure ...	3m
Q 2	∠COB = _____°	G1: Use angle properties ...	2m
Q 3	∠AOC = _____°	G1: Use angle properties ...	2m
Q 4	∠AOD = _____°, ∠BOD = _____°, ∠BOC ...	G1: Use angle properties ...	3m
Q 5	∠C = _____°	G1: Apply angle sum of tr...	2m
Q 6	Perpendicular: _____, Parallel: ___...	G1: Identify perpendicula...	2m
Q 7	∠COD = _____°	G1: Combined angle proper...	3m
Q 8	∠XOY = _____°	G1: Measure reflex angles	2m
Q 9	Perimeter = _____ cm	G2: Calculate perimeter o...	2m
Q10	Perimeter = _____ cm	G2: Calculate perimeter o...	3m
Q11	Area = _____ cm^2	G2: Calculate area of com...	3m
Q12	Remaining area = _____ cm^2	G2: Calculate area with m...	3m
Q13	(a) _____, (b) _____, (c) _____	G2: Convert between units...	3m
Q14	2.5 m^2 = _____ cm^2	G2: Convert between units...	2m
Q15	Number of lines = _____	G4: Identify lines of sym...	2m
Q16	(Complete the drawing on the diagra...	G4: Complete symmetrical ...	3m
Q17	Volume = _____ cm^3	G3: Calculate volume of c...	2m
Q18	Volume = _____ cm^3	G3: Calculate volume with...	3m
Q19	Valid net(s): _____	G3: Identify valid nets o...	2m
Q20	(Draw on the grid provided)	G3: Draw nets of cuboids	4m
Q21	Type: _____ triangle	G4: Classify triangles by...	1m
Q22	Type: _____ triangle	G4: Classify triangles by...	1m
Q23	Quadrilateral: _____	G4: Classify quadrilatera...	2m
Q24	Circumference = _____ cm	G2: Calculate circumferen...	2m
Q25	Fraction = _____	G2: Interpret pie charts ...	3m